Coherence of Meridional Heat Transport in the Atlantic

Drivers of variability and change..

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Lea Sophie Dietz, July 2026

Outline - Content

1. What is interesting about heat transport?
   1.1. Why do we look at it ? Global heat distribution etc
   1.2. Connection with AMOC here already - AMOC weakening?
2. What are recent observations? What is the research gap?
   2.1. Continous measurements since 2000s (RAPID) what remains unknown ?
3. Objectives
4. Data & Methods
   4.1. How is MHT measured?
   4.2. Dataset that I use - Calafat , BHM
5. First results / ideas
6. Outlook - what remains to be done

Motivation - Research Gap

Why do we care about heat transport in the Atlantic Ocean?

- Generally: how does heat transport work globally?
  - Global energy budget
  - What is special about the atlantic?
    - Deep OHC changes, connection with Arctic - through changes in the AMOC
    - AMOC responsible for most heat (and carbon) transport in NH
- Model say: reduction in OHT , predict weakening of AMOC in this century

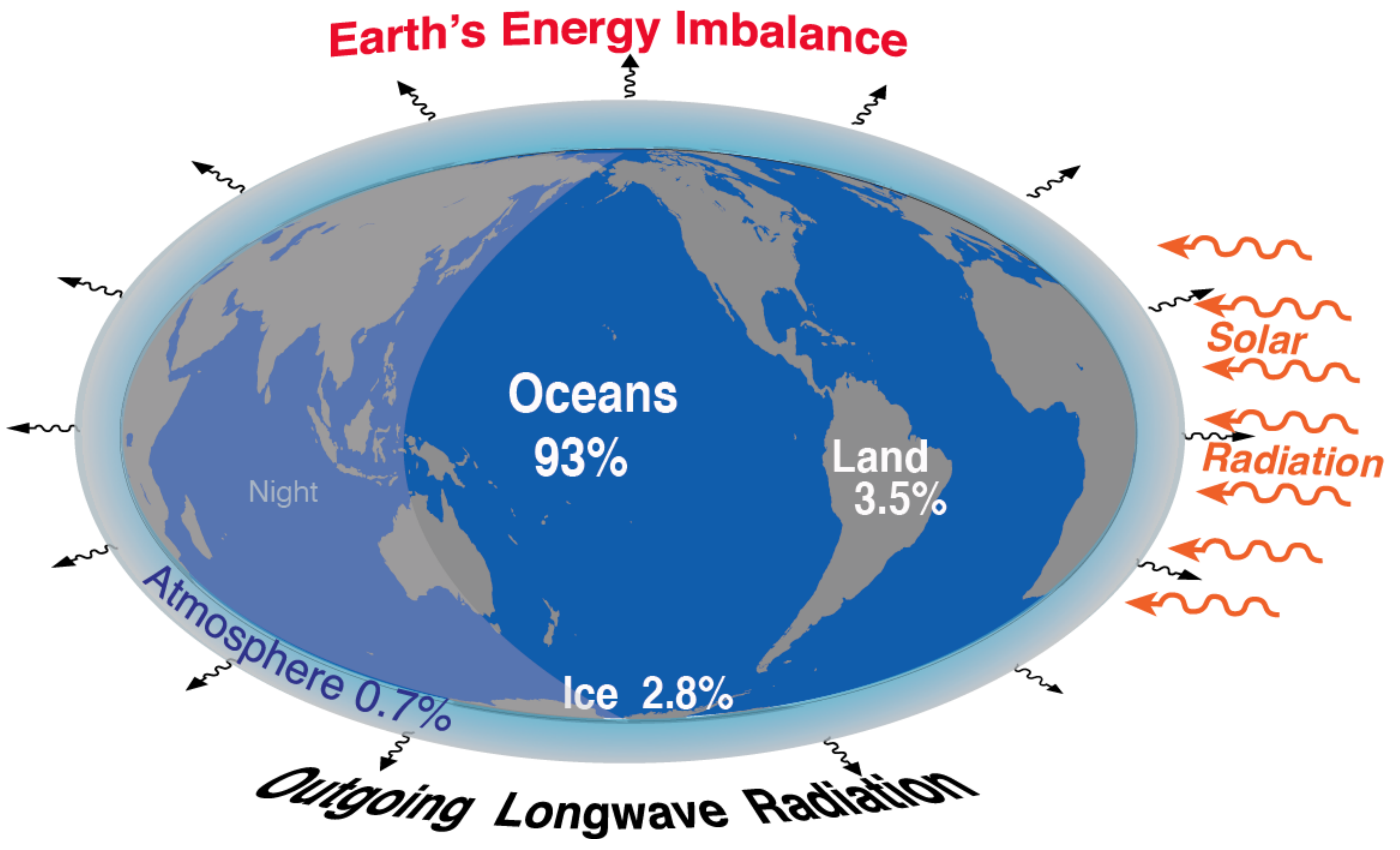
- Variability: important to understand what is inherent variability vs. what is long-term human induced trend, need to now how signals propagate and change

Motivation - Research Gap

Why do we care about heat transport in the Atlantic Ocean?

- Perspective: Earth Energy Budget and Imbalance
  - Imbalance: currently more energy absorbed than emitted
  - Ocean is dominant heat sink with an uptake of 93% of this imbalance energy

→ This excess heat enters the ocean via surface heat flux



Wikipedia

Motivation - Research Gap

Why do we care about heat transport in the Atlantic Ocean?

- Surface heat flux F_S - energy (heat) is absorbed in the ocean
- This heat then is either stored or recirculated

$$F_s = -dOHC/dt - \nabla \cdot \mathbf{F_O}$$

dOHC/dt: how fast is heat stored / released, >0 ⇒ Ocean warming

∇F_O: divergence OHC (transport of ocean energy) ⇒ $F_O = F_{circulation} + F_{gyre} + F_{eddies}$

Rearranging for ∇F_O gives an equation for MHT from the energy budget:

$$\mathrm{MHT}(\phi) = \int_{\phi}^{90} [F_s + dOHC/dt]\, a d\phi$$

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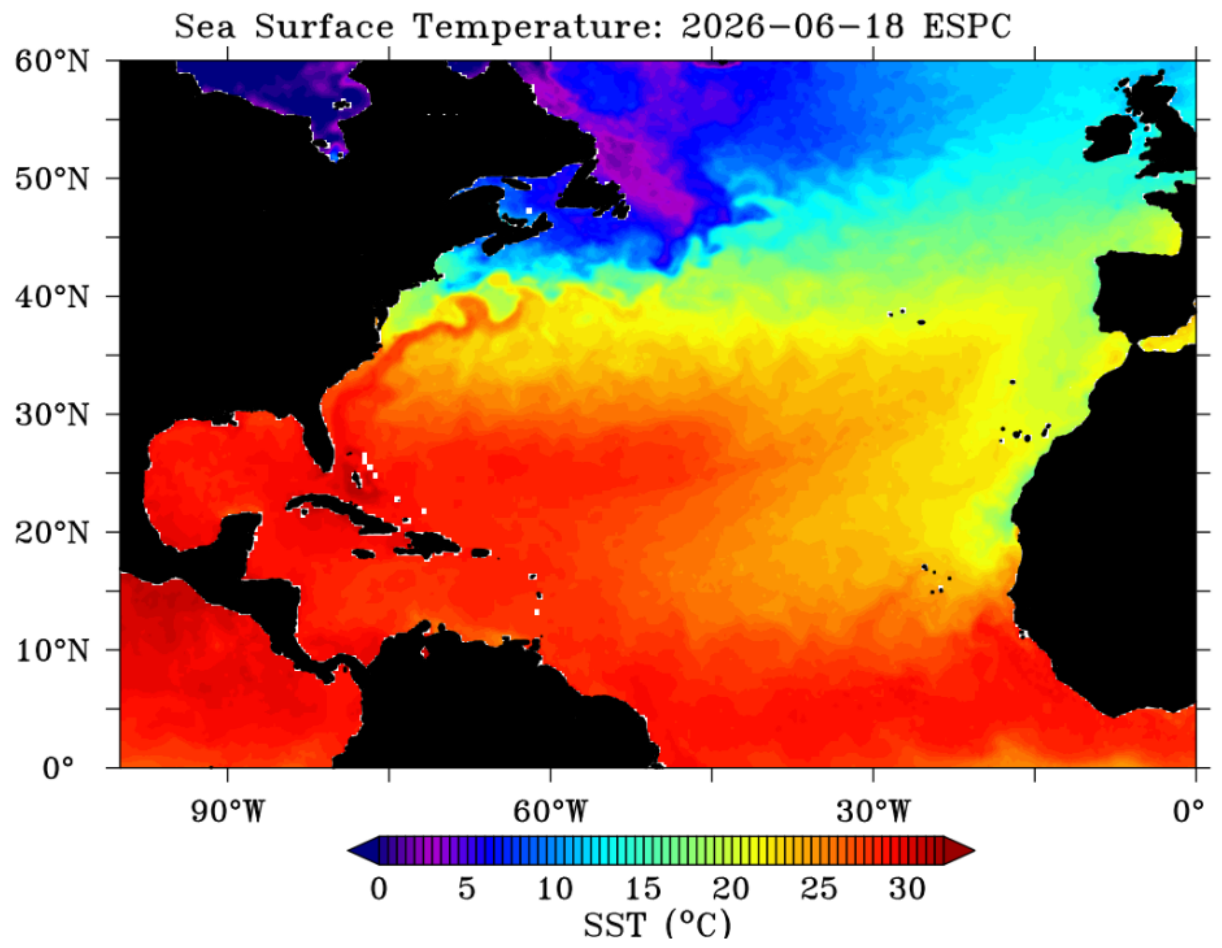
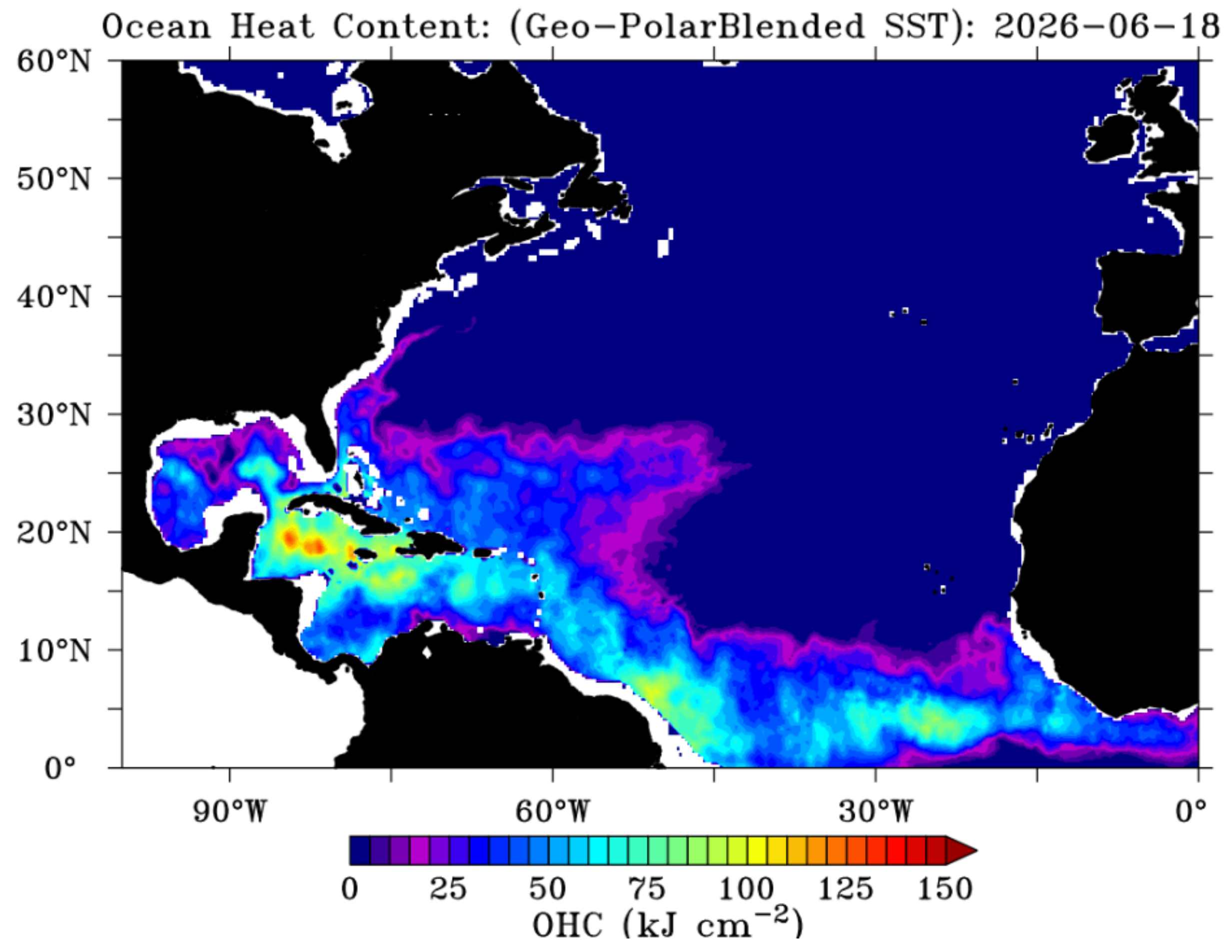
Motivation - Research Gap

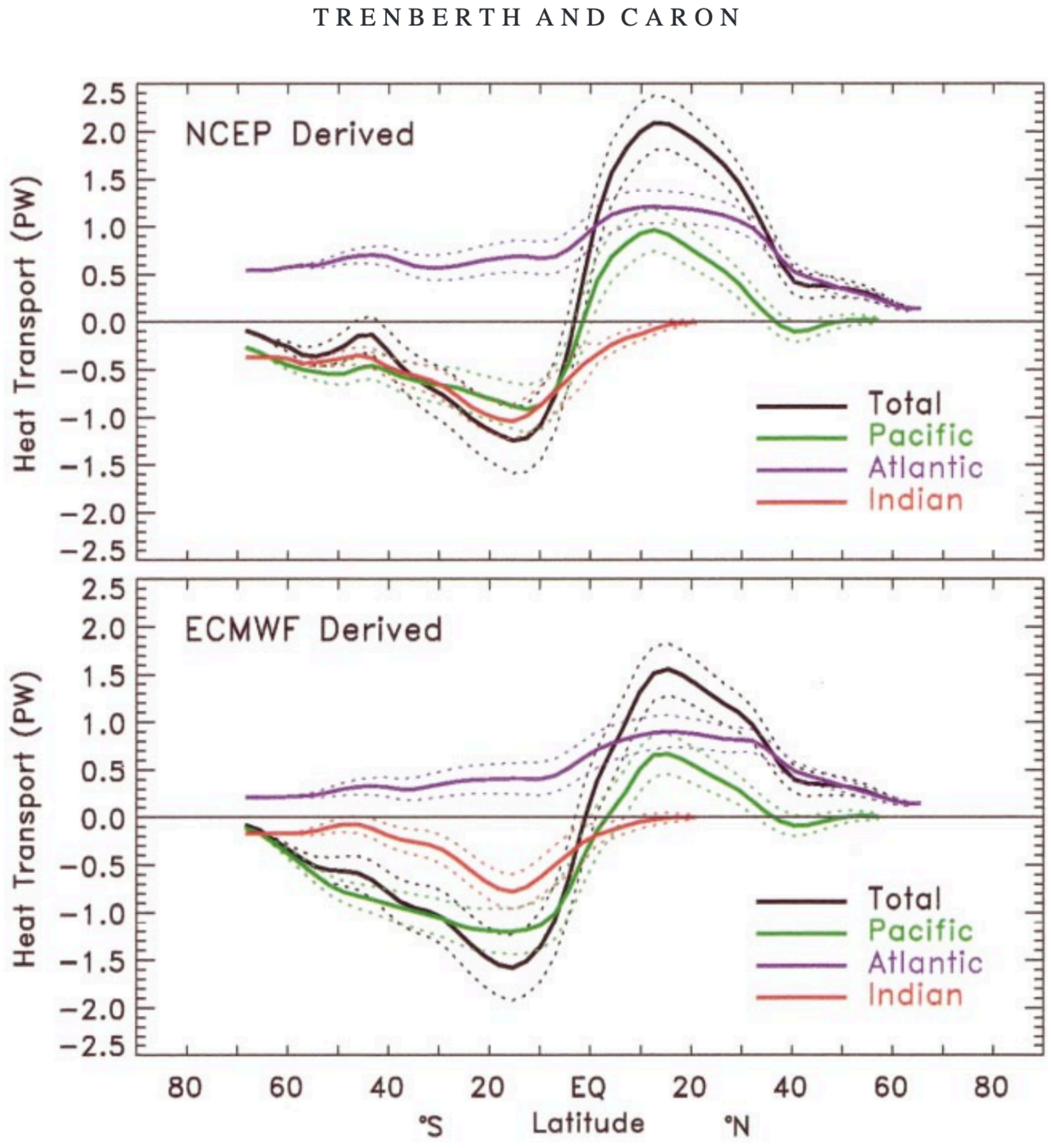
Why do we care about heat transport in the Atlantic Ocean?

- Highest heat input close to equator - high SST and OHC
- Heat transport in Atlantic only place where is solely northward





https://www.ospo.noaa.gov/products/ocean/ohc/



Trenberth and Caron (2001)

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Motivation - Research Gap

Why do we care about heat transport in the Atlantic Ocean?

Next: to understand what drives changes in MHT: $\Delta(\mathrm{MHT}) = \Delta(\mathrm{OHC})/\Delta t + \Delta(\mathrm{ASHF}).$

Motivation - Research Gap

Why do we care about heat transport in the Atlantic Ocean?

- AMOC transport = heat transport (mainly) 0.065 PW /Sv or something like this
- AMOC weaken - less heat northward- change in climate all over (not only Europe)
  - Position of ITCZ connects with heat transport over equator - change in ITCZ can have drastic impacts on equator near areas
- Less heat transported away from equator
- Important to understand what drives changes in MHT
- How does it propagate through the basin? Highest heat input at the Equator
- Not everywhere fully coherent
- If not coherent- due to changes in air-sea heat flux or changes in circulation
- Eq for MHT ?

Motivation - Research Gap

Why do we care about heat transport in the Atlantic Ocean?

- Data needed - continuous obs range (as rapid) only since 2004
- Longest continuous measurement is RAPID 20+ years of observation since 2004 for basin wide measurements
  - Satellite already earlier used
  - Use calafat data which combines- show it estimates MHT comparable to other datasets
  - Fig 8 from calafat plot?

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Motivation - Research Gap

What is meridional heat transport?

$$Q_{\mathrm{NET}} = \int\int \rho c_p v\theta \, dx \, dz,$$

$$\Delta(\mathrm{MHT}) = \Delta(\mathrm{OHC})/\Delta t + \Delta(\mathrm{ASHF}).$$

Depends on

θ, potential temperature - air sea heat flux

v, meridional velocity - which strongly depends on AMOC!

At 26°N the ocean (AMOC) carries 1.3 PW of heat northwards - 70% of global net poleward heat flux (Johns et al. 2011)

When MHT changes so does the storage of heat change in the ocean

- less heat transported - heat stays near equator - oceans capacity to store is maximized - atmosphere heats even more

Bryden et al. (2020)

Methods - Data

How can we measure MHT?

- Direct in-situ measurements (Moorings etc)
- Downside:
  - Expensive, restricted to few latitudes (i.e. RAPID or OSNAP)
  - Spatially sparse and uneven distribution of data can introduce bias and errors

→ Idea: combine hydrographic data with satellite for probabilistic MHT estimates

Using Bayesian hierachical model

MHT as residual

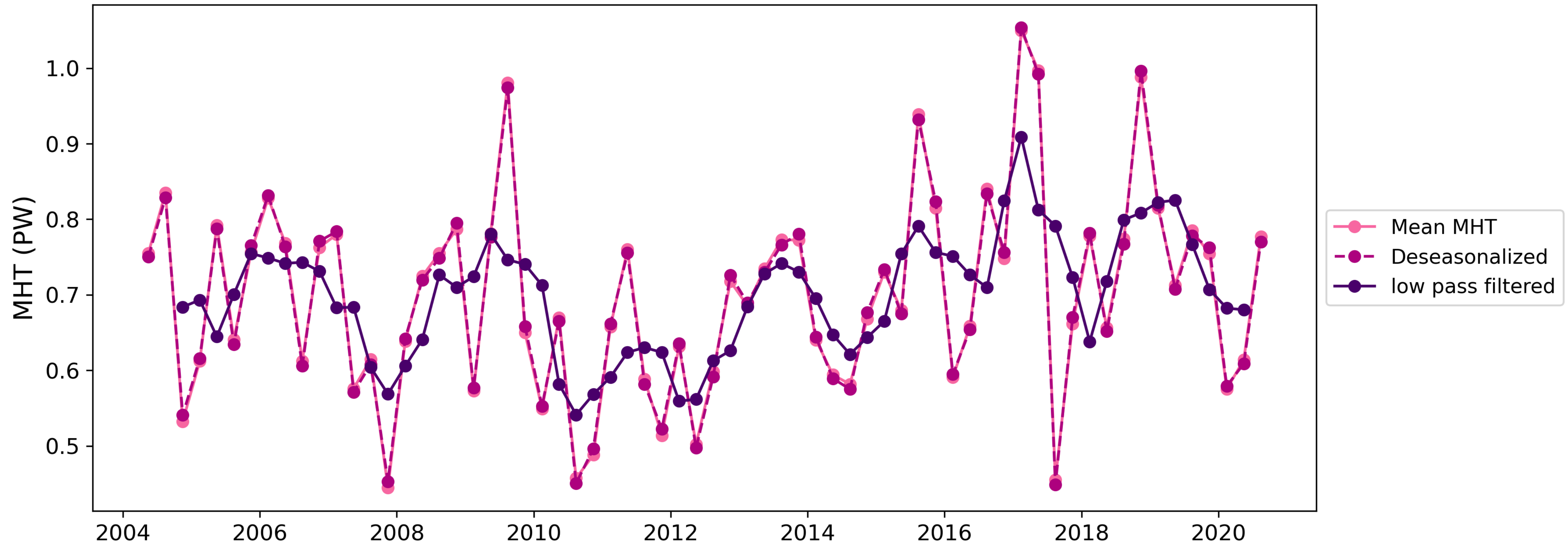
Thesis objectives

Investigate coherence of MHT and drivers for incoherence

1. How coherent is MHT over the atlantic?
2. What drives incoherence? Are differences caused by changes in surface heat flux or by circulation changes?

Using the MHT estimates from BHM (*Calafat et al. 2025)*

Processing by: removing seasonal cycle + 1 year low pass filter → isolate interannual variability



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Was able to reproduce finding from calafat for two covarying bands

Using MHT anomalies relative to each latitude band

Cross correlation shows significant correlation for bands:

35 - 16°N and **5°N - 35°S**